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SUCC - Sequence and UMI Clustering-based Correction

Presented by Mohit Patil and Nikolaos Mitsopoulos on 18.03.2026

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Motivation

- PCR-Amplification
  - False expression
  - Unnecessary workload for read mapper
- Sequencing Errors
  - False differentiation between unique molecules

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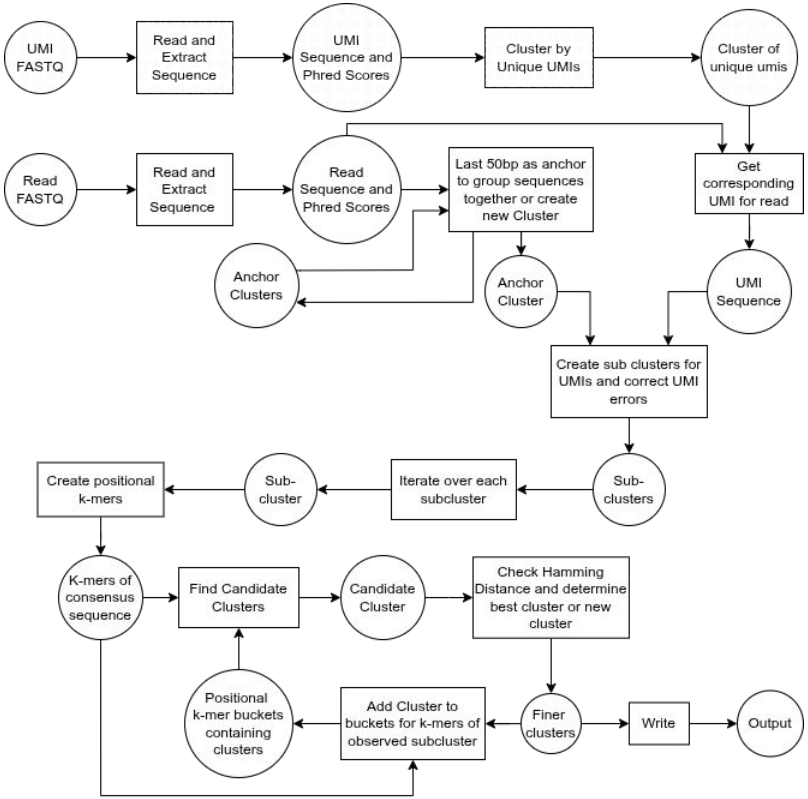
Workflow



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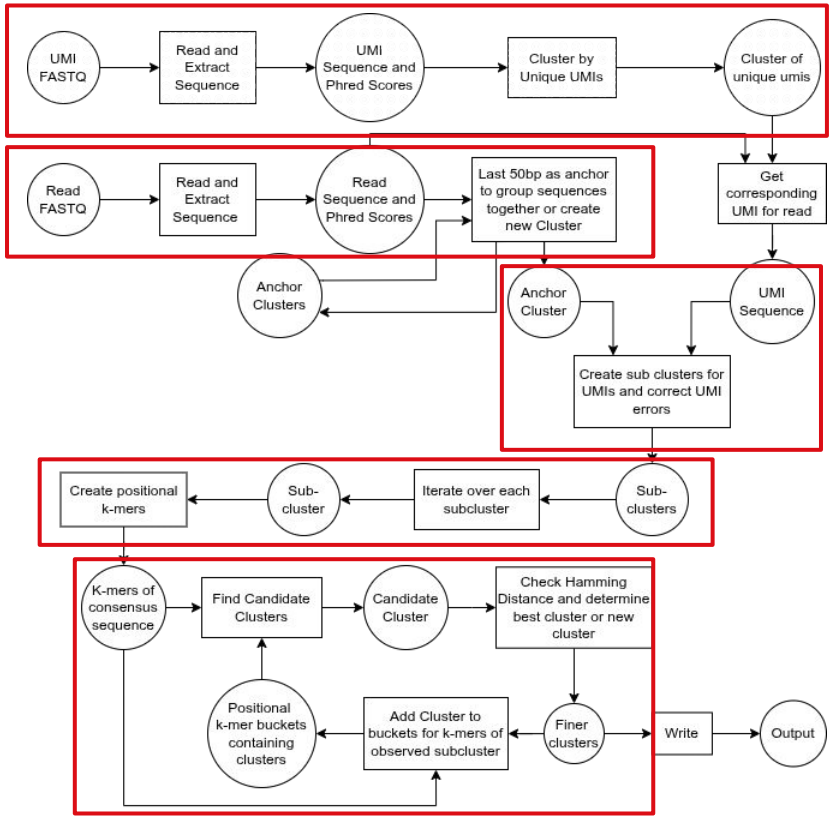
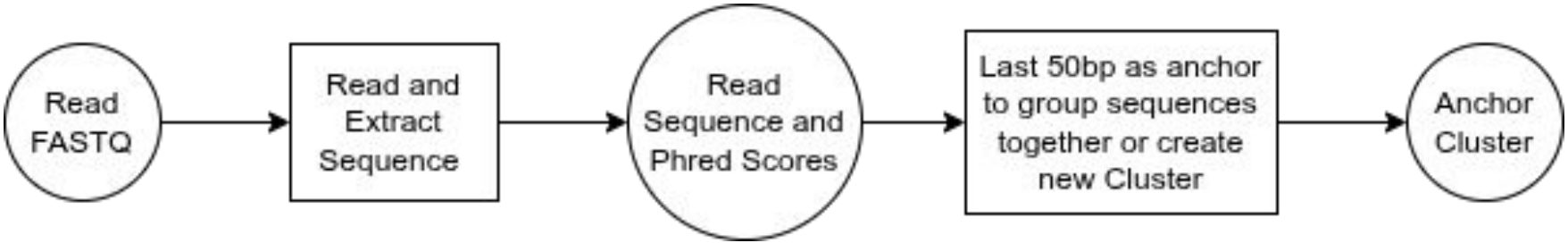
Workflow



UMI-Clustering

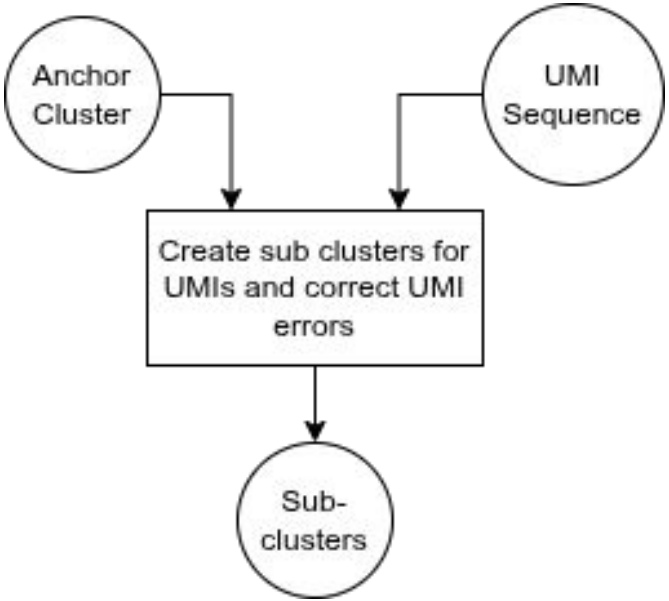
Anchor Clustering



UMI-Correction and subcluster construction

- Correction of nucleotides with a bad quality
- Phred-score =<20
- Allowing errors at up to 2 positions

```
ATCTAGTCCTCC
+
IIII9II9**II
```



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Building Consensus String

- Naive approach based on the assumption, that errors are a minority.
- Phred score is accumulated for a similar sequence.
- Replacement of bad quality bases with better quality ones.
- Rare for UMIs → bad quality positions are mutated to merge onto cluster
- Problematic when:
  - Different Fragments with same sequence in last 50 nucleotides
  - Identical UMI / erroneous UMI identical after repairing.

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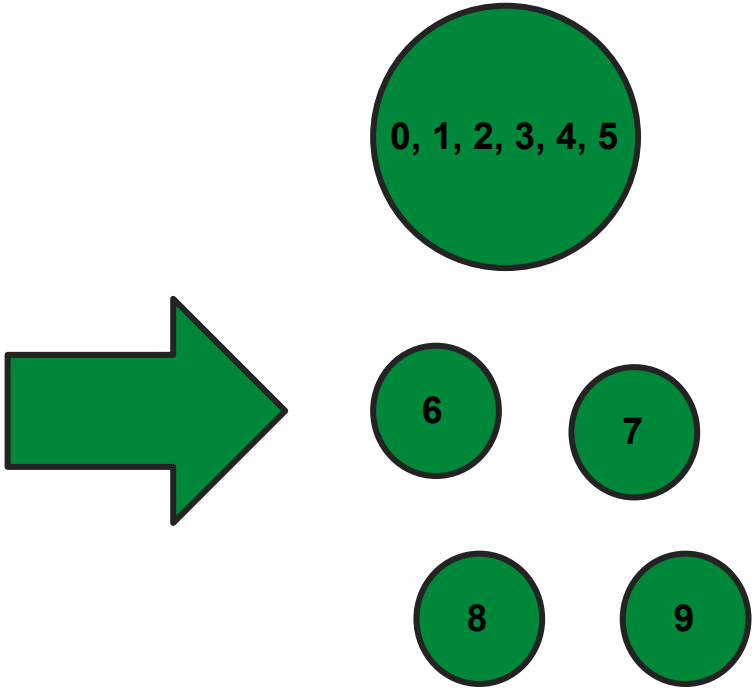
Finer clustering: Motivation

- Each resulting cluster is defined by its seeds last 50 bp and UMI sequence.
  - Errors in the last 50 bp lead to under-clustering

- Probability of observing at least one error in a 50 bp sequence:
  - $(0.99)^{50} \approx 40\%$

Expected consequences

ID	UMI	Sequence
0	ATC	GT…ATCGATCGATCCCCG
1	ATC	GT…ATCGATCGATCCCCG
2	ATC	GT…ATCGATCGATCCCCG
3	ATC	GT…ATCGATCGATCCCCG
4	ATC	GT…ATCGATCGATCCCCG
5	ATC	GT…ATCGATCGATCCCCG
6	ATC	GT…ATCGATCGCTCCCCG
7	ATC	GT…ATCGATCGATCCCTG
8	ATC	GT…ATCGGTCGATCCCCG
9	ATC	GT…ATCGATCGATCACCG



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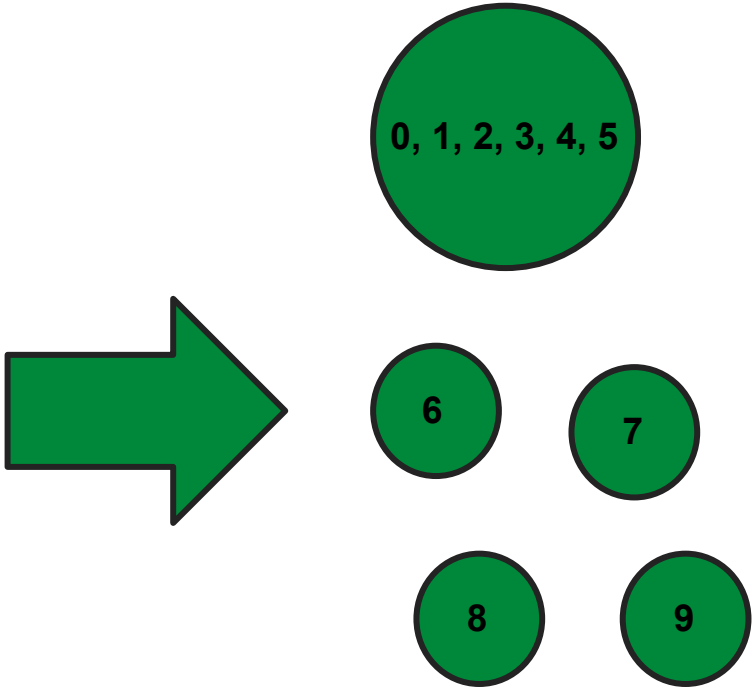
5	ATC	GT…ATCGATCGATCCCCG
6	ATC	GT…ATCGATCGCTCCCCG
7	ATC	GT…ATCGATCGATCCCTG
8	ATC	GT…ATCGGTCGATCCCCG
9	ATC	GT…ATCGATCGATCACCG



10M reads, ~10× PCR duplication

⇒ ~4M extra clusters

Implementation

Greedy clustering approach:

Instead of computing all pairwise distances, each incoming read is compared only to existing clusters.

A cluster is represented by a **seed sequence**.

Anchor strategy: the first encountered sequence defines the seed of a cluster.

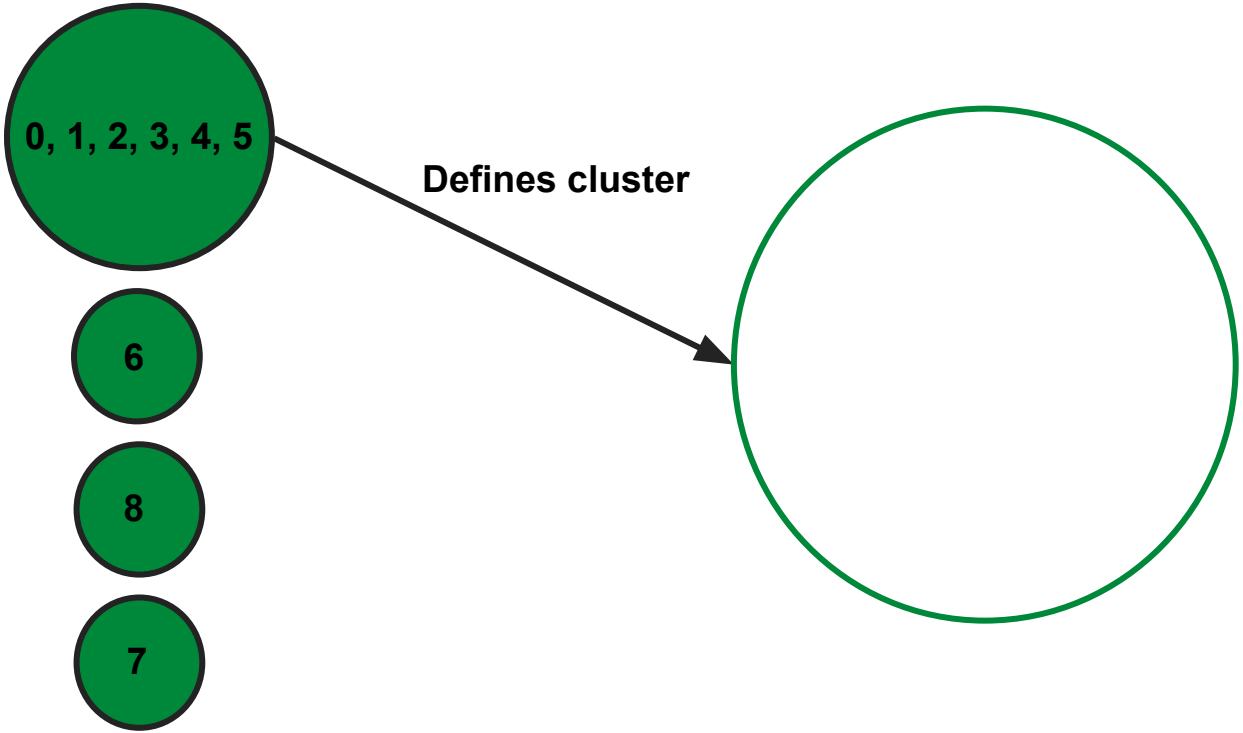
Anchor strategy: Weakness

Anchor strategy: the first encountered sequence defines the seed of a cluster.

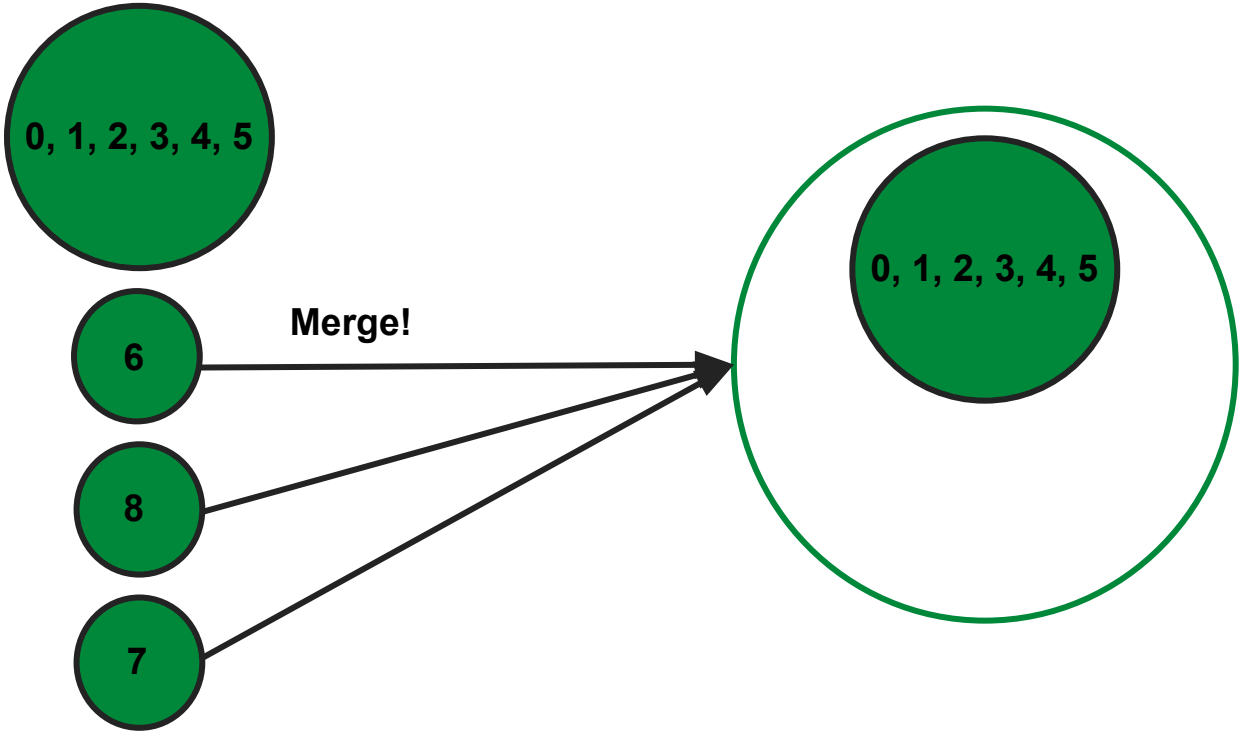
⇒ Clustering quality depends on the read order.

Solution: Sort before clustering!

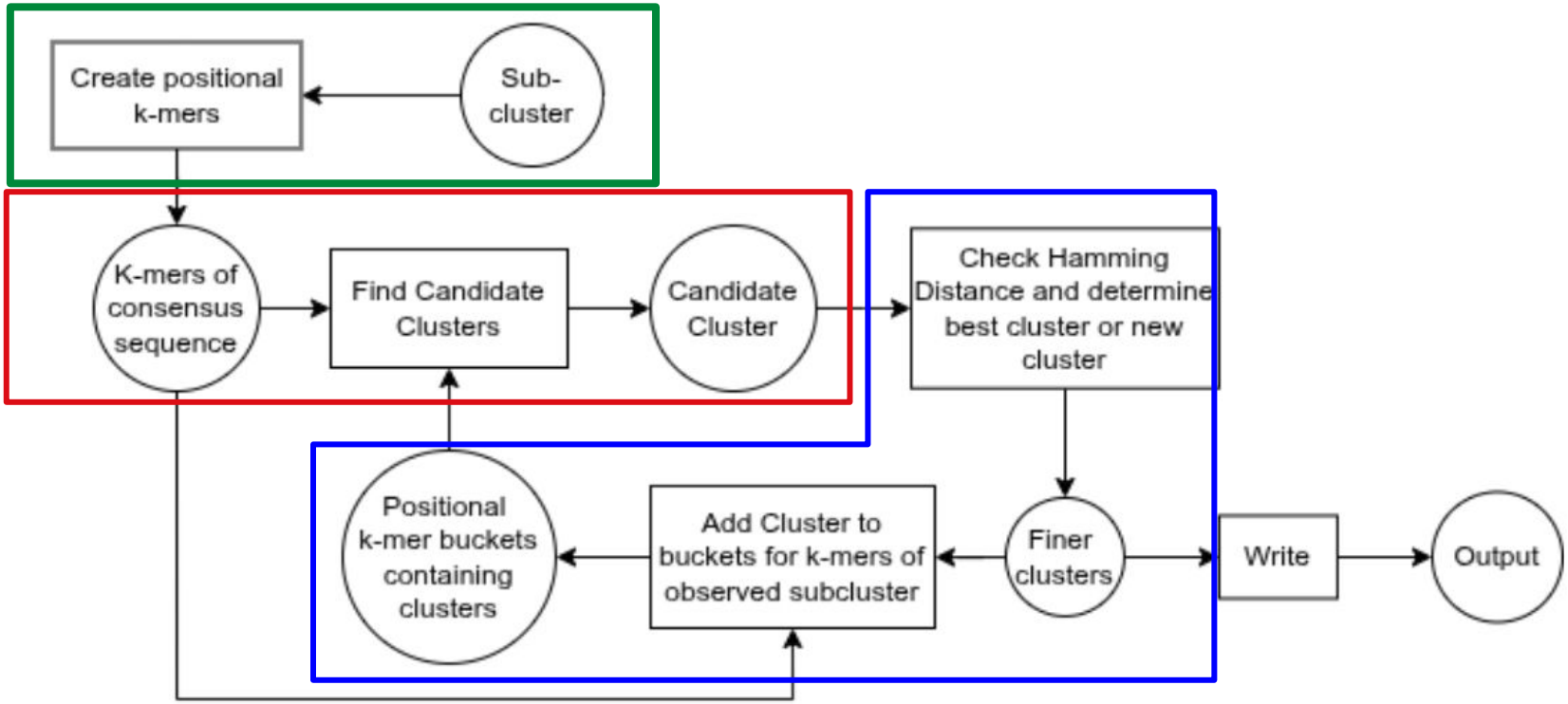
Expected consequences



Expected consequences



Finer clustering



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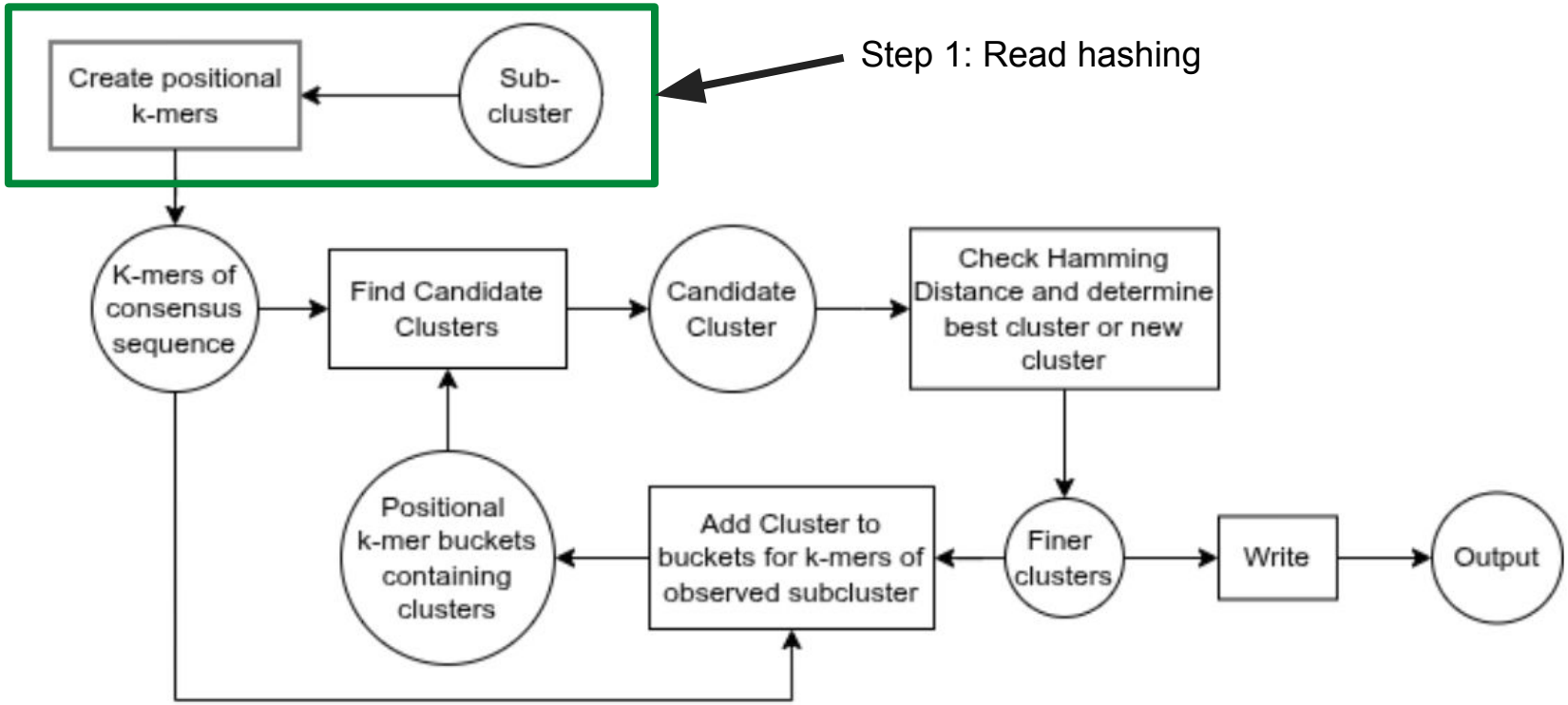


Read hashing: Positional k-mers

ATCGGGCTAATCGATCGATCCCCGGTTAAAGCGACTGGCTAGCTAGCTGA

Split

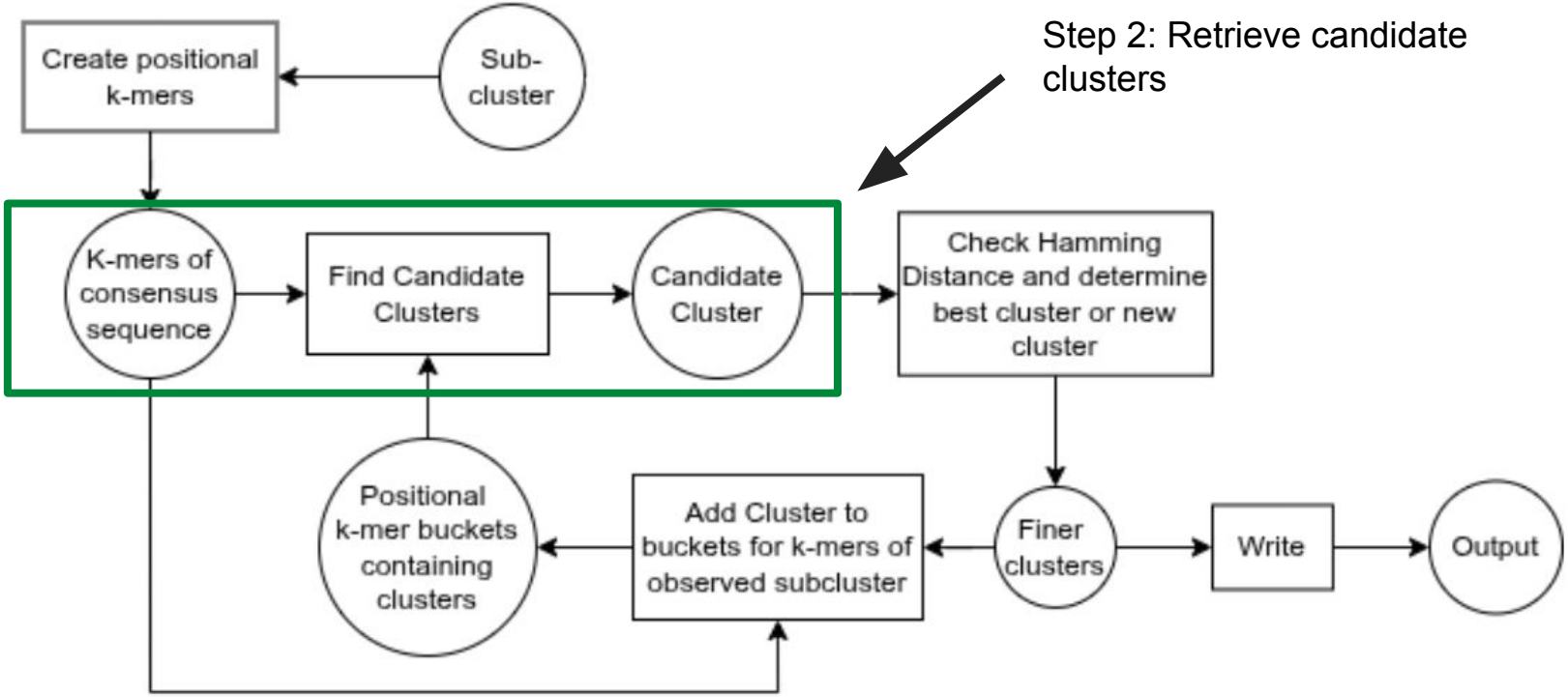
ATCGGGCTAA **TCGATCGATC** **CCCGGTTAAA** **GCGACTGGCT** **AGCTAGCTGA**

Encode

00011011111110010000
01101100011011000110
01010111111010000000
11101100100111111001
00111001001110011100

ordered

Finer clustering



Candidate cluster search

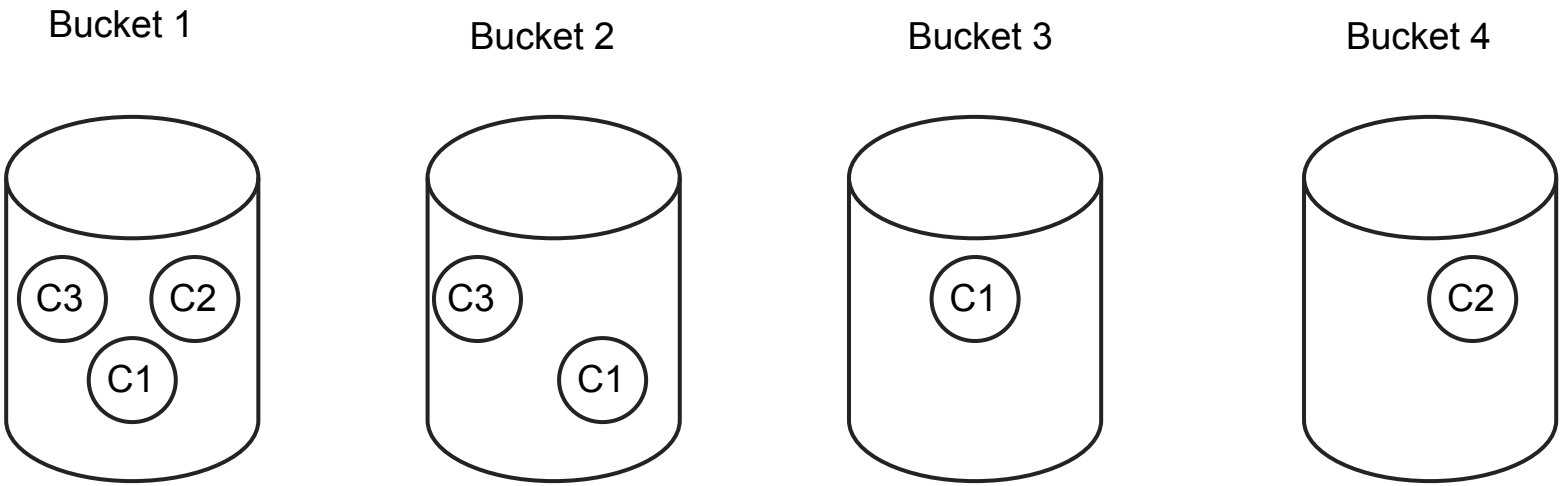
Why searching all clusters is expensive

- Comparing each new read against all existing clusters is computationally costly
- As the number of clusters grows, the search space increases continuously

Observation in real data

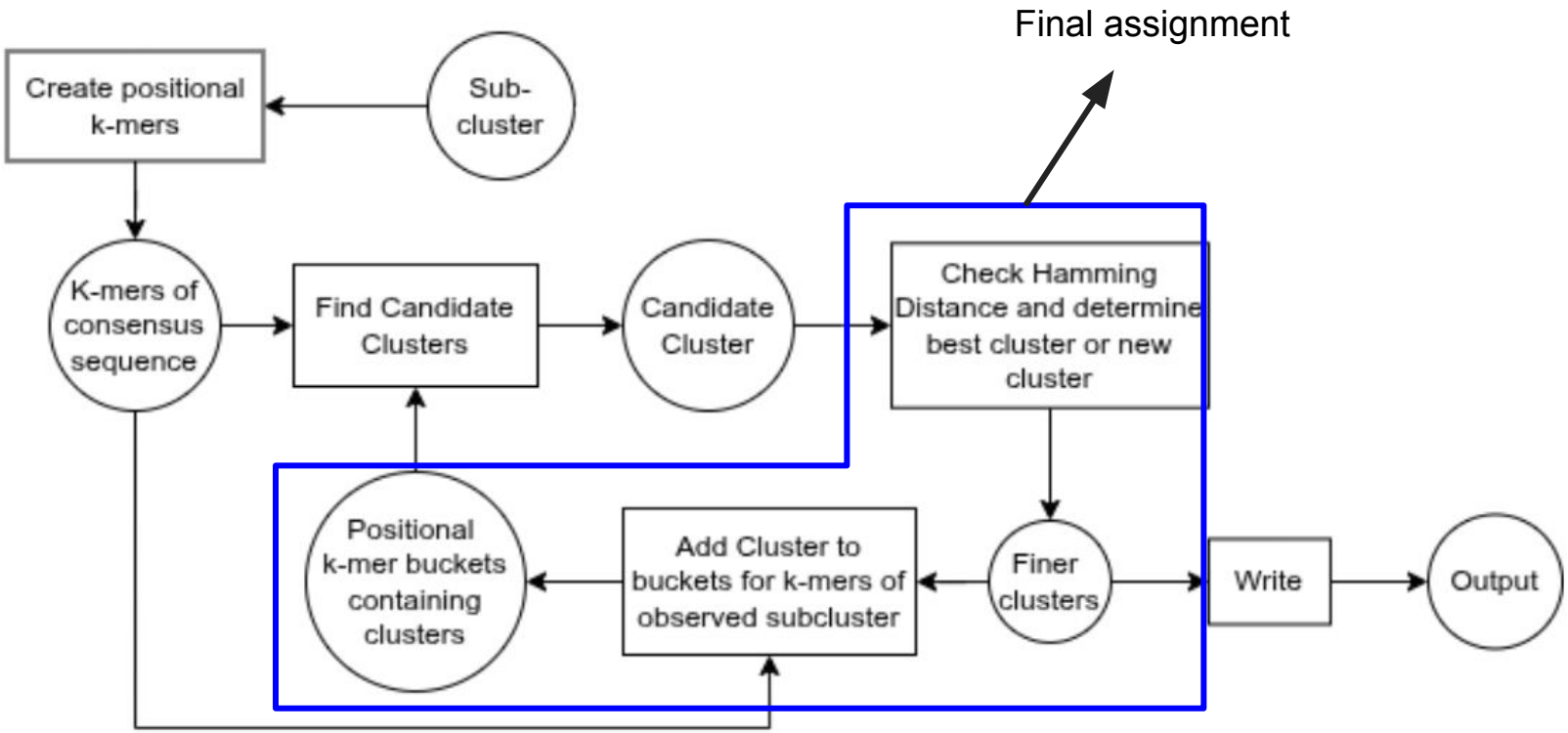
- Most clusters are singletons
- Many comparisons are spent checking clusters that are unlikely to attract additional reads

Solution: Cluster buckets



- Assign each cluster to buckets using the seeds hash value
- Use hash value of query read to retrieve relevant clusters

Finer clustering



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Final assignment

- Compute hamming distance of query read against seed of all candidate clusters.
- If minimal distance out of all candidates is smaller than the threshold
  - Assign to existing cluster
- Else create new cluster

- Out of multiple metrics: Hamming distance yielded the best results

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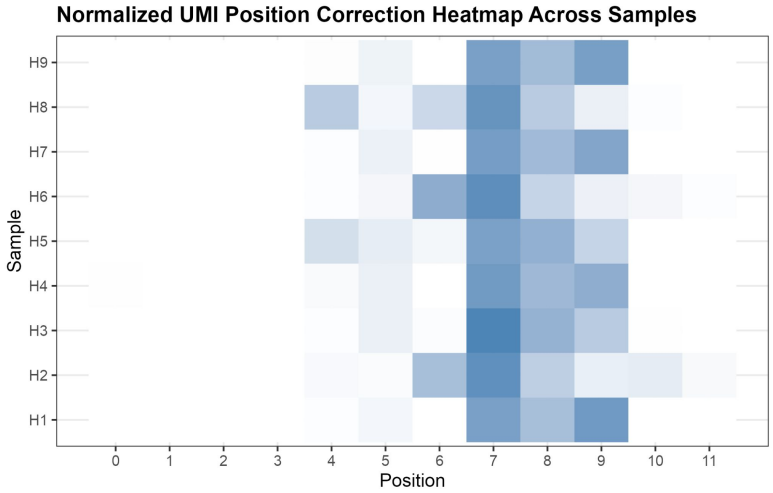
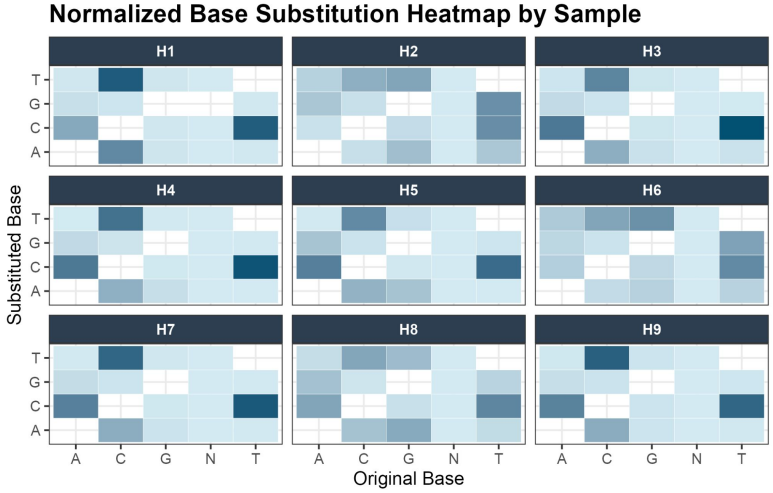
UMI Correction Results

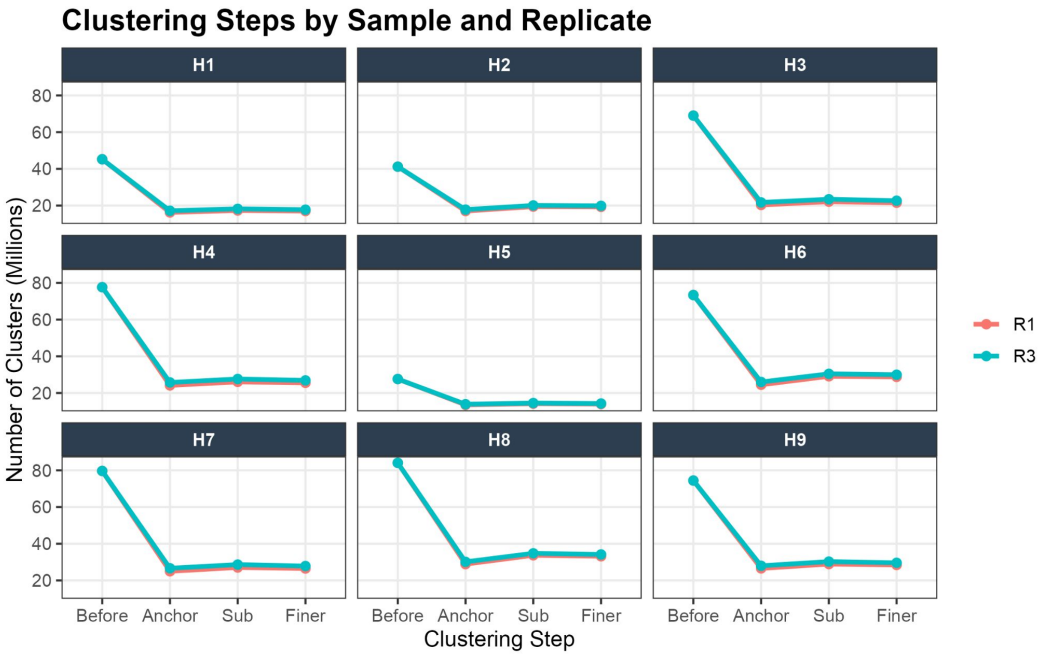
Sample	Before correction	After correction R1	After correction R3
H1	8551	8366	8316
H2	20023	19811	19677
H3	19322	19066	18877
H4	11880	11650	11526
H5	6991	6875	6822
H6	21081	20885	20685
H7	11411	11198	11099
H8	18009	17780	17672
H9	10325	10135	10067



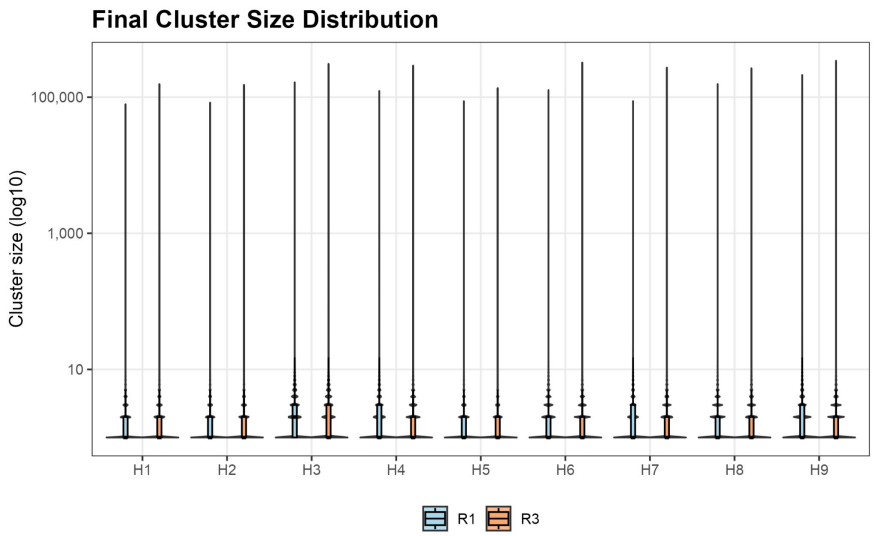


Amount of Deduplication

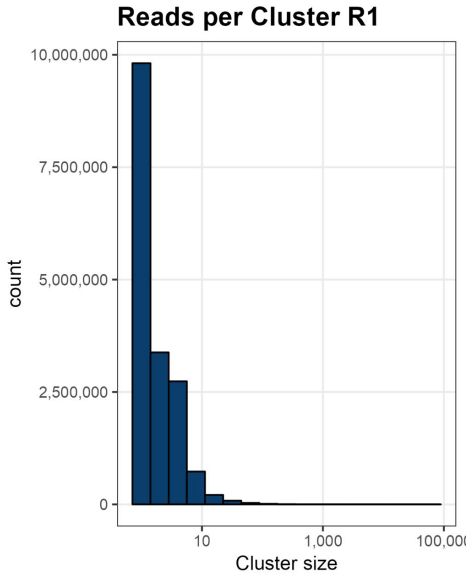
- Number of reads halved
- Differences between R1 and R2, e.g. H1:
  - R1: 16.98 million clusters
  - R2: 17.70 million clusters

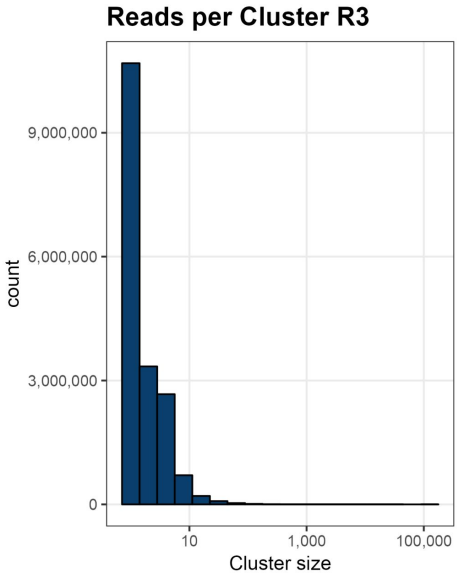


Cluster sizes



H1:





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Evaluation framework

- **Simulation Setup:**
- Sus scrofa dataset
  - Extract transcript counts (bamfeature analyzer)
  - Simulate expression (modified read simulator)

- Defining ground truth
  - Two reads should be clustered together if and only if:
  - The originate from the same genomic position
  - They share the same UMI

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Simulation results

Total reads	19,998,423
Read length	150 bp
UMI length	12 bp
Sequencing error rate	1%
PCR cycles	18
PCR efficiency	85%
Unique fragments	6,174,255
Unique UMIs	8,500

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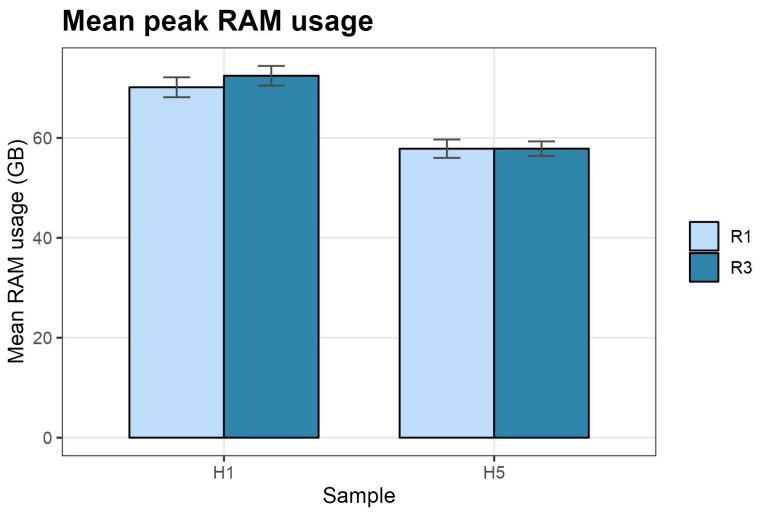
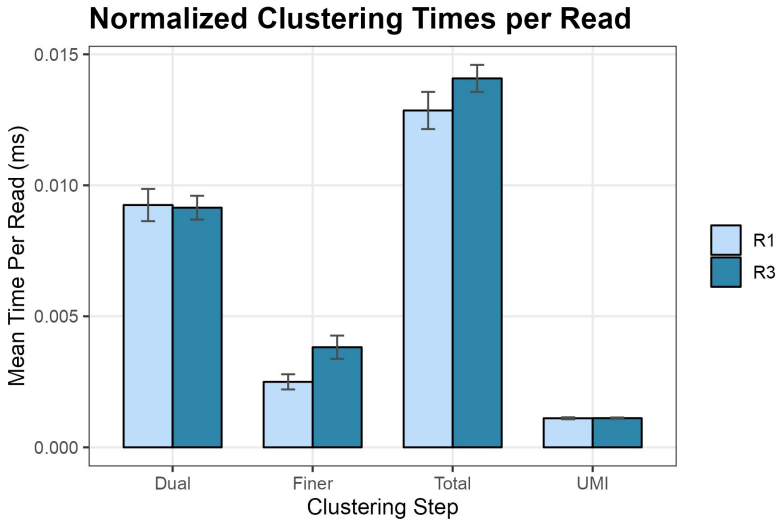
Simulation results

Metric	Primary only	Secondary only	Both
Pred. Clusters	13,290,236	4,505,486	6,312,607
ARI	0.5374	0.3922	0.9945
Precision	0.9999	0.2467	0.9999
Recall	0.3674	0.9562	0.9891
F1	0.5374	0.3922	0.9945
Homog.	0.9999	0.9601	0.9999
Compl.	0.9540	0.9970	0.9955
	Under-clustering	Over-clustering	Just right

Benchmarks

Sample	Read type	Mean Time
H1	R1	9 min 30 s
H1	R3	10 min 50 s
H5	R1	6 min 2 s
H5	R3	6 min 20 s





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Conclusion

- Two-cycle clustering approach demonstrated good performance
  - Near-perfect quality on simulated data
  - ~50% reduction of reads on Sus. Scrofa dataset

- Real data insights:
  - PCR amplification biases and “jackpot” instances impact clustering
  - Forward and reverse read yield different number of clusters

- Optimal parameters are dataset-dependent
- Careful tuning is required for real word applications

Thank you for listening!!!

Questions?